

Mathematical Foundations for Machine Learning

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1. Solution of Linear Systems

Linear systems are one of the first places where machine learning becomes matrix mathematics. A dataset can place one observation in each row, a model parameter in each column, and a target value on the right. The compact equation $Ax = b$ then asks whether one parameter vector x can satisfy all observations at once.

[Detailed External Sources](#)

From Equations to a Matrix

Consider the equations

$$x + 2y = 5$$

$$3x - y = 4$$

They can be written as

$$A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix} \quad x = \begin{bmatrix} x \\ y \end{bmatrix} \quad b = \begin{bmatrix} 5 \\ 4 \end{bmatrix} \quad Ax = b$$

- A is the coefficient matrix.
- x is the vector of unknowns.
- b is the right-hand-side vector.
- $[A \mid b]$ is the augmented matrix, which attaches b as the last column.

If A has m rows and n columns, then A maps vectors in $\mathbb{R}^n$ to vectors in $\mathbb{R}^m$. Matrix-vector multiplication is defined because the n entries of x match the n columns of A .

Matrix Operations That Matter

For matrices of the same shape, addition is entry by entry. Multiplication by a scalar scales every entry. Matrix multiplication is different: if A is m by n and B is n by p , then AB is m by p , with

$$(AB)_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$$

The order matters. In general, AB is not equal to BA , and one product may exist when the other does not.

The transpose A^T exchanges rows and columns. Useful rules include

- $(A^T)^T = A$
- $(A + B)^T = A^T + B^T$

- $(AB)^T = B^T A^T$

The reversal in the last rule is important.

Dimension check before calculation

If A is m by n , then x must contain n entries and b must contain m entries. The product Ax combines the n unknowns into m equation values. A dimension mismatch means the proposed multiplication is undefined.

With one row per observation and one column per feature, a machine-learning design matrix with m observations and n features multiplies an n -parameter vector to produce m predictions.

Elementary Row Operations

Three operations preserve the solution set of a linear system:

- 1 Swap two rows.
- 2 Multiply a row by a nonzero scalar.
- 3 Add a multiple of one row to another row.

These operations replace equations with equivalent equations. Applied systematically, they produce an echelon form.

In row echelon form (REF):

- every nonzero row is above every zero row;
- each leading entry, or pivot, is to the right of the pivot above it;
- entries below each pivot are zero.

Reduced row echelon form (RREF) adds two conditions:

- every pivot equals 1;
- every pivot is the only nonzero entry in its column.

REF is usually enough for back substitution. RREF makes the solution structure immediately visible and is unique for a given matrix.

Worked Example: One Solution

Solve

$$x + 2y = 5$$

$$3x - y = 4$$

Start with the augmented matrix:

$$\left[\begin{array}{cc|c} 1 & 2 & 5 \\ 3 & -1 & 4 \end{array} \right]$$

Apply $R_2 \leftarrow R_2 - 3R_1$:

$$\left[\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & -7 & -11 \end{array} \right]$$

The second row gives $y = 11/7$. Substituting into the first equation gives

$$x = 5 - 2 \left(\frac{11}{7} \right) = \frac{13}{7}$$

So the unique solution is $x = 13/7$, $y = 11/7$. A direct check gives both original equations exactly.

Rank and Consistency

The rank of a matrix is the number of pivot positions in an echelon form. It measures how many independent row or column directions the matrix contains.

For a system with n unknowns:

- If $\text{rank}(A) < \text{rank}([A \mid b])$, the system is inconsistent and has no solution.
- If $\text{rank}(A) = \text{rank}([A \mid b]) = n$, the system has exactly one solution.
- If $\text{rank}(A) = \text{rank}([A \mid b]) < n$, the system has infinitely many solutions, with $n - \text{rank}(A)$ free variables.

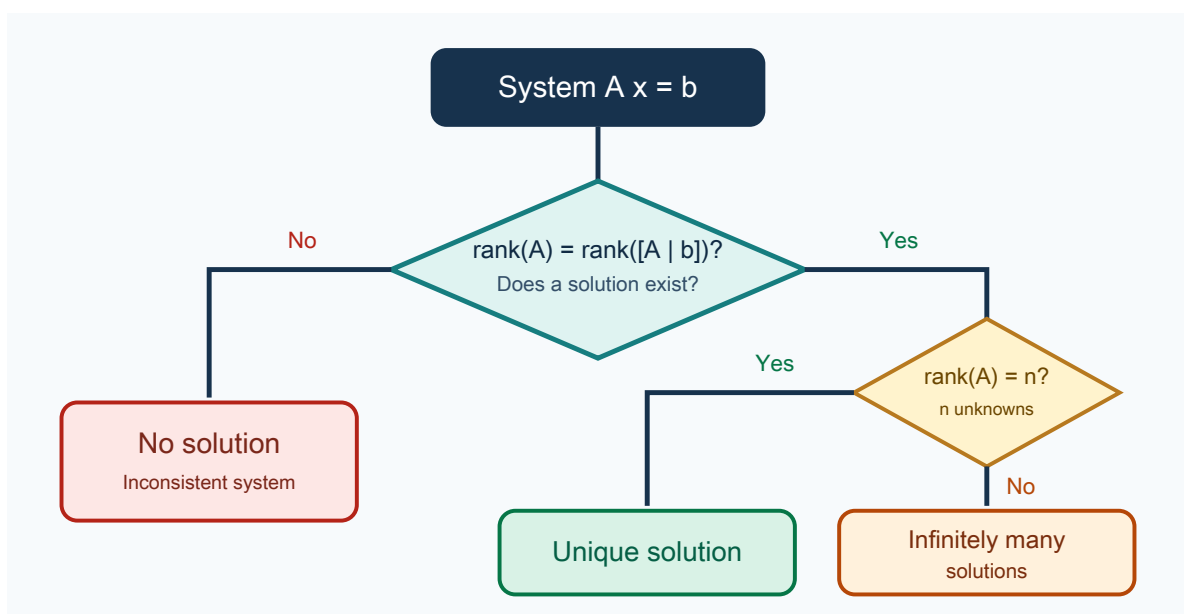


Figure 1. Rank separates inconsistent, uniquely solvable, and underdetermined consistent systems.

Seeing the Three Outcomes

No Solution

$x + y = 1$ and $x + y = 3$ have the same left side but different right sides. Elimination produces a row of the form

$$0x + 0y = 2$$
$$\Rightarrow 0 = 2$$

This contradiction makes the system inconsistent.

Infinitely Many Solutions

$x + y = 2$ and $2x + 2y = 4$ describe the same line. One row reduces to zero, leaving one pivot and one free variable. Writing $y = t$ gives

$$x = 2 - t, \quad \text{and} \quad \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Unique Solution

Two independent equations in two unknowns produce two pivots. Geometrically, two nonparallel lines meet at one point.

Homogeneous Systems

A homogeneous system has the form $Ax = 0$. It always has the trivial solution $x = 0$.

If A has fewer pivots than unknowns, at least one free variable exists and the system also has nontrivial solutions. Therefore,

$Ax = 0$ has a nontrivial solution

$$\Leftrightarrow \text{rank}(A) < n$$

The complete solution set of a homogeneous system is the null space of A , a subspace discussed in [Know More](#).

Numerical Practice

For a square, full-rank matrix, numerical software can solve $Ax = b$ directly. Forming A^{-1} first is usually unnecessary and can be less accurate and more expensive.

The condition number estimates sensitivity: a large condition number means small input or rounding changes can produce much larger changes in the computed solution. This is distinct from exact singularity. A matrix may be invertible in exact arithmetic yet numerically difficult to use.

Numerical libraries provide direct solvers for square, full-rank systems. Non-square or rank-deficient systems require a method suited to least squares or underdetermined equations. Numerical rank is tolerance-dependent because floating-point singular values are rarely exactly zero.

What Singularity Actually Means

- A square matrix with determinant zero has no inverse.
- For a particular $Ax = b$, a singular matrix may give no solution or infinitely many solutions.
- The outcome depends on whether b belongs to the column space of A .

Real Application: Mixing Ingredients

Suppose a food producer blends two ingredients. Each kilogram of ingredient 1 contributes 3 units of protein and 1 unit of fibre; ingredient 2 contributes 1 unit of protein and 2 units of fibre. To obtain 13 protein units and 8 fibre units, the unknown ingredient amounts satisfy

$$3x + y = 13$$

$$x + 2y = 8$$

Elimination gives $y = 11/5$ and $x = 18/5$. The matrix is not just shorthand: its columns describe what one unit of each ingredient contributes. If those columns were proportional, the blend could not independently control protein and fibre, and some nutritional targets would be impossible.

Checking a Computed Solution

Substitute the candidate into the original equations, or calculate the residual

$$r = b - Ax^{\wedge}$$

where $\hat{x}$ is the computed solution. In exact hand arithmetic, a correct solution gives $r = 0$. In floating-point computation, a very small residual is expected because decimal arithmetic is rounded. A small residual confirms that the values nearly satisfy the equations, but an ill-conditioned system can still make the solution sensitive to tiny input changes.

Common Mistakes

- Multiplying matrices entry by entry instead of using row-by-column products.
- Forgetting that $(AB)^T = B^T A^T$ reverses order.
- Applying an operation to A but not to the matching entry in b .
- Counting nonzero entries instead of pivots when finding rank.
- Declaring a system inconsistent merely because it has fewer pivots than unknowns.
- Treating a nearly singular floating-point matrix as if exact symbolic rules guaranteed a stable answer.

Takeaway

Row reduction reveals the structure hidden in $Ax = b$. Rank first decides whether the equations are compatible; when they are, the number of pivots decides whether the solution is unique or contains free directions.

[Detailed](#) [Know More](#) [External Sources](#)

2. Vector Spaces

A vector space tells us which objects may be added and scaled without leaving the world we are studying. This matters in machine learning because feature vectors, parameter vectors, images stored as arrays, and solution sets of homogeneous equations can all be treated with the same rules.

[Detailed](#) [External Sources](#)

Operations and Scalar Fields

A binary operation combines two elements of a set and returns another element of the same set. Addition on real numbers is one example.

A group is a set with one binary operation satisfying closure, associativity, an identity element, and an inverse for every element. If the operation is also commutative, it is an abelian group.

A field supplies the scalars used to scale vectors. The real numbers $\mathbb{R}$ are the usual field in this course. The field rules make addition, subtraction, multiplication, and division by nonzero values behave consistently.

What Is a Vector Space?

A vector space V over a field F has two operations:

- vector addition: $u + v$;
- scalar multiplication: $c u$, where c is in F .

For all vectors u, v, w in V and scalars c, d in F , these operations obey:

- 1 $u + v$ is in V .
- 2 $c u$ is in V .
- 3 $(u + v) + w = u + (v + w)$.
- 4 There is a zero vector 0 with $u + 0 = u$.
- 5 Every u has an additive inverse $-u$ with $u + (-u) = 0$.
- 6 $u + v = v + u$.
- 7 $c(u + v) = cu + cv$.
- 8 $(c + d)u = cu + du$.
- 9 $c(du) = (cd)u$.
- 10 $1u = u$, where 1 is the multiplicative identity in F .

Rules 1 and 3-6 say that vectors form an abelian group under addition. Rule 2 and the remaining rules describe how field scalars act within the space and interact with vector addition.

Linear Combinations: The Core Operation

A linear combination of vectors $v_1, v_2, \dots, v_k$ is

$$c_1 v_1 + c_2 v_2 + \dots + c_k v_k$$

where $c_1, c_2, \dots, c_k$ are scalars. Vector spaces are designed so that every finite linear combination remains inside the space.

This idea connects the chapter: matrix-vector multiplication forms a linear combination of the columns of A ; a subspace is closed under linear combinations; and $Ax = b$ is solvable exactly when b can be formed from the columns of A .

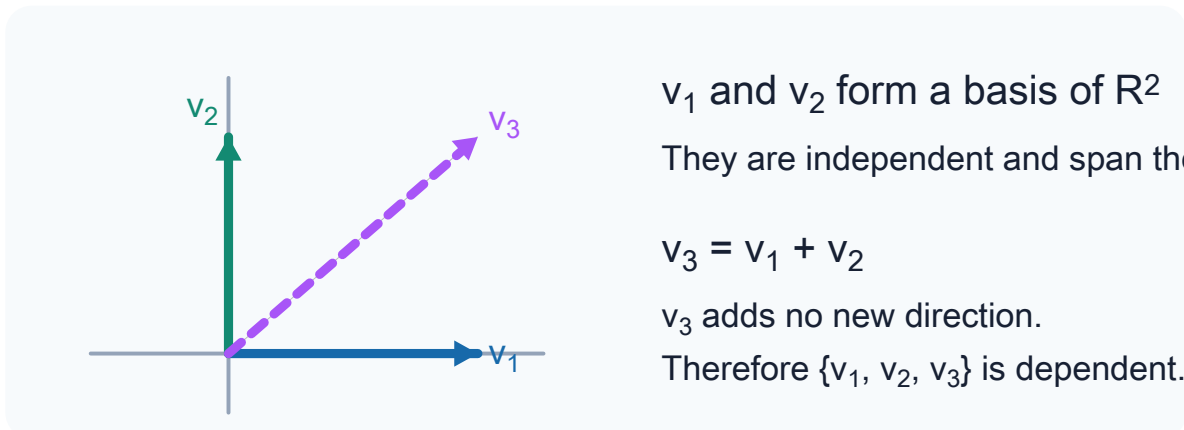
Span, Linear Independence, and Basis

The span of a list of vectors is the set of every linear combination that can be built from them. The list spans a space when those combinations reach every vector in that space.

A list is linearly independent when the zero vector can be formed only by setting every coefficient to zero. If a nonzero choice of coefficients also produces zero, the list is linearly dependent: at least one vector is redundant because it can be written using the others.

$$c_1 v_1 + c_2 v_2 + \dots + c_k v_k = 0$$

A basis is a list that is both linearly independent and spanning. It is therefore a nonredundant coordinate system for the space. The number of vectors in any basis is the space's dimension.



In the figure, v_1 and v_2 form a basis of $\mathbb{R}^2$. Adding $v_3 = v_1 + v_2$ does not enlarge their span, so the three-vector list is dependent. This illustrates an important distinction: a dependent list may still span the whole space; it simply contains redundancy.

For a matrix, row reduction gives a reliable test. Put the candidate vectors into columns. A pivot in every column means they are independent; a pivot in every row means their columns span the whole output space. For a square matrix, either condition implies the other.

Examples

Coordinate Vectors

$\mathbb{R}^n$ is the set of all ordered lists of n real numbers. Addition and scalar multiplication happen component by component. It is the standard finite-dimensional vector space used for numeric features.

For example, daily sales across three regions can be represented by the following explicit column vectors. Their sum is another vector in $\mathbb{R}^3$.

$$u = \begin{bmatrix} 12 \\ 8 \\ 5 \end{bmatrix} \quad v = \begin{bmatrix} 3 \\ 6 \\ 4 \end{bmatrix} \quad u + v = \begin{bmatrix} 15 \\ 14 \\ 9 \end{bmatrix}$$

Matrices

The set of all m by n real matrices is also a vector space. Adding two matrices of the same shape or scaling every entry preserves that shape.

This matters when a dataset itself is treated as one point in a matrix space, such as a grayscale image represented by a fixed-size pixel matrix.

Subspaces

A subspace U of V is a subset that is itself a vector space using the same operations as V .

Most vector-space axioms are inherited automatically. A practical test is enough:

- 1 0 belongs to U .
- 2 If u and v belong to U , then $u + v$ belongs to U .
- 3 If u belongs to U and c is a scalar, then cu belongs to U .

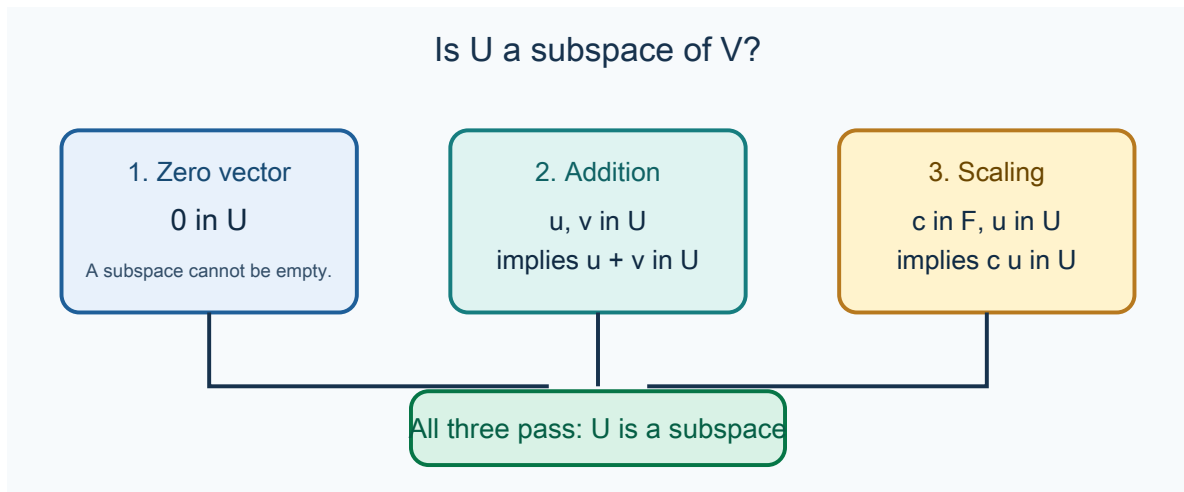


Figure 1. A candidate subspace must contain zero and be closed under addition and scaling.

The last two checks can be combined: for all scalars a, b and all u, v in U , the vector $au + bv$ must also belong to U .

Worked Subspace Check

Let

$$U = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 : x + y + z = 0 \right\}$$

Zero: The explicit zero column vector satisfies the equation.

$$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Addition: If $x_1 + y_1 + z_1 = 0$ and $x_2 + y_2 + z_2 = 0$, then

$$(x_1 + x_2) + (y_1 + y_2) + (z_1 + z_2) = 0$$

Scaling: If $x + y + z = 0$, then for any real c ,

$$cx + cy + cz = c(x + y + z) = 0$$

All three checks pass, so U is a subspace of $\mathbb{R}^3$.

Now compare

$$S = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2 : y = 2 \right\}$$

The zero vector is not in S , so S is not a subspace. Geometrically, it is a line shifted away from the origin.

Fast counterexample strategy

To prove that a set is not a subspace, one failed condition is enough. Check the zero vector first. If zero belongs, look for two members whose sum leaves the set or one member whose scalar multiple leaves it. To prove that a set is a subspace, however, every required condition must hold.

Important Subspaces of a Matrix

For an m by n matrix A :

- the column space contains all vectors Ax and lies in $\mathbb{R}^m$;
- the row space is the span of the rows and lies in $\mathbb{R}^n$;

- the null space contains all x satisfying $Ax = 0$ and lies in $\mathbb{R}^n$.
- the left null space contains all y satisfying $A^T y = 0$ and lies in $\mathbb{R}^m$.

All four are subspaces. The equation $Ax = b$ is consistent exactly when b lies in the column space of A . The row and column spaces have the same dimension, namely $\text{rank}(A)$.

For an m by n matrix of rank r , their dimensions are:

Fundamental subspace	Ambient space	Dimension
Column space	$\mathbb{R}^m$	r
Row space	$\mathbb{R}^n$	r
Null space	$\mathbb{R}^n$	$n - r$
Left null space	$\mathbb{R}^m$	$m - r$

The two rank-nullity relationships account for every input and output direction:

$$\begin{aligned}\text{rank}(A) + \text{nullity}(A) &= n \\ \text{rank}(A^T) + \text{nullity}(A^T) &= m\end{aligned}$$

Rank as Dimension Information

The rank of A is the number of pivots in an echelon form. It is also the dimension of the column space and the row space.

For an m by n matrix,

$$\text{rank}(A) \leq \min(m, n)$$

Elementary row operations preserve rank. Transposition also preserves rank:

$$\text{rank}(A^T) = \text{rank}(A)$$

In a homogeneous system, the number of free variables is

$$n - \text{rank}(A)$$

This quantity is the dimension of the null space. It explains why fewer pivots than unknowns produce nontrivial homogeneous solutions.

Exact Rank and Numerical Rank

On paper, rank is an exact algebraic property. In floating-point computation, very small singular values must be compared with a tolerance. Therefore, numerical software may classify a nearly dependent direction as zero even when an exact symbolic matrix has full rank.

This is not a contradiction: the numerical question is whether a direction is distinguishable at the available precision.

Real Application: Valid Feature Transformations

Imagine that each vector in $\mathbb{R}^3$ stores the red, green, and blue intensity of a pixel. All possible colour vectors form a vector space: two colour vectors can be added and a vector can be scaled, although a display may later clip values to its legal range.

Now consider the set of colour adjustments whose three channel changes sum to zero. This set is a subspace because adding two such adjustments or scaling one preserves the zero-sum condition. By contrast, all colours whose channels sum to 1 form a shifted plane. That plane misses the zero vector, so it is not a subspace.

This distinction appears in machine learning whenever a constraint is homogeneous. Solutions of $Ax = 0$ form a subspace; solutions of $Ax = b$ for nonzero b , when they exist, generally form a shifted version of the null space rather than a subspace.

Common Mistakes

- Calling any subset of $\mathbb{R}^n$ a subspace.
- Checking closure under addition but forgetting scalar multiplication.
- Forgetting that every subspace must contain the zero vector.
- Assuming a shifted line or plane is a subspace even when it misses the origin.
- Confusing the column space, which lies in $\mathbb{R}^m$, with the null space, which lies in $\mathbb{R}^n$.
- Calling a spanning list a basis without checking independence.
- Assuming that more vectors always create a larger span; redundant vectors do not.
- Testing independence with a determinant when the vector matrix is not square. Row reduction works for rectangular matrices too.
- Treating a numerical rank result as independent of tolerance.

Takeaway

A vector space is not defined by arrows or coordinates; it is defined by operations and rules. Subspaces preserve those rules inside a smaller set. A basis is a nonredundant spanning list, and matrix rank records how many independent row and column directions survive.

Before accepting a proposed subspace, ask whether it contains zero and is closed under addition and scalar multiplication. Before interpreting $Ax = b$, remember that Ax is a linear combination of columns and rank measures how many independent output directions those columns provide.

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4. Matrix Decomposition - I

The determinant compresses one important property of a square matrix into a single number. It tells us whether a linear transformation collapses a dimension, how oriented area or volume is scaled, and whether a square system can have a unique solution for every right-hand side.

[Detailed](#) [External Sources](#)

Determinant of a 2 by 2 Matrix

For

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

the determinant is

$$\det(A) = ad - bc$$

Example:

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$$

gives

$$\det(A) = 3(2) - 1(1) = 5$$

Because the determinant is nonzero, A is invertible.

Minors, Cofactors, and Expansion

For a square matrix, the minor M_{ij} is the determinant left after deleting row i and column j .

The corresponding cofactor is

$$C_{ij} = (-1)^{i+j} M_{ij}$$

The signs alternate in a checkerboard pattern:

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

Expanding a 3 by 3 matrix along the first row gives

$$\det(A) = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$$

The same determinant results from expansion along any row or column. A row or column containing many zeros is usually the easiest choice.

Worked 3 by 3 Example

Let

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 1 \\ -1 & 0 & 3 \end{bmatrix}$$

Expand along the first row:

$$\det(A) = 1 \det \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} + 1 \det \begin{bmatrix} 0 & 2 \\ -1 & 0 \end{bmatrix}$$

Therefore,

$$\det(A) = 1(6) + 1(2) = 8$$

A direct row-reduction or another cofactor expansion gives the same result.

Geometric Meaning

In two dimensions, the columns of A form the sides of a parallelogram. Its area is $|\det(A)|$ when the original unit square has area 1.

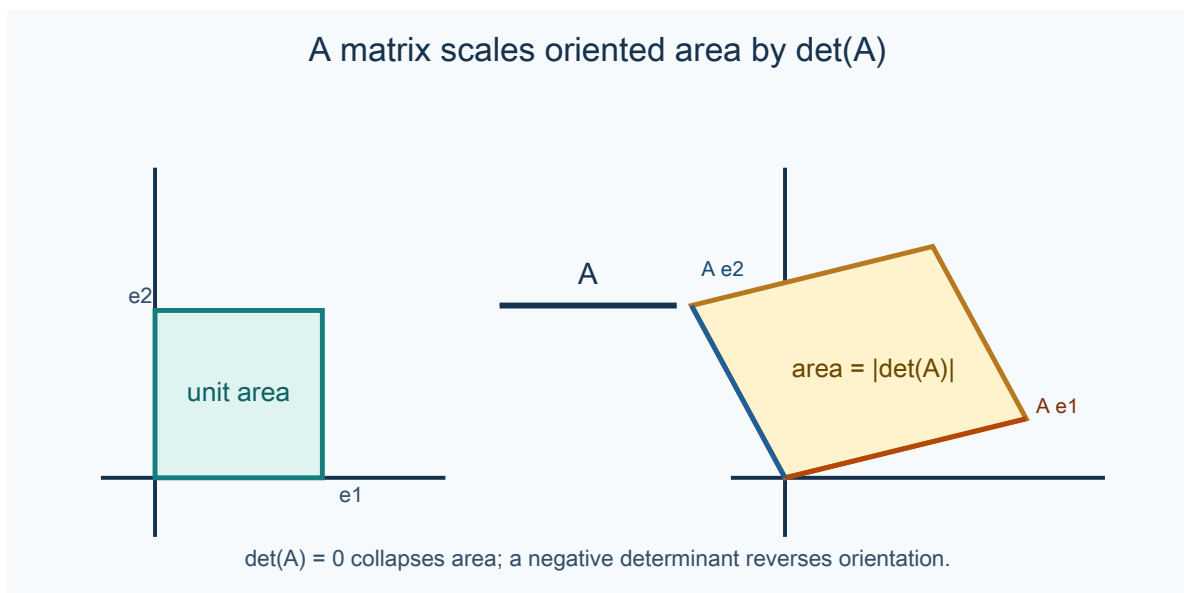


Figure 1. The determinant is a signed area or volume scale factor.

- $|\det(A)| > 1$ expands area or volume.
- $0 < |\det(A)| < 1$ contracts it.
- $\det(A) = 0$ collapses at least one dimension.
- $\det(A) < 0$ reverses orientation in addition to scaling.

This geometric view explains why a zero determinant means the columns cannot span the full space.

Determinant Properties

For square matrices of the same size:

- $\det(AB) = \det(A)\det(B)$.
- $\det(A^T) = \det(A)$.
- Swapping two rows changes the determinant's sign.
- Multiplying one row by c multiplies the determinant by c .
- Adding a multiple of one row to another row leaves the determinant unchanged.
- A triangular matrix has determinant equal to the product of its diagonal entries.
- Equal or linearly dependent rows force the determinant to zero.

These rules often make elimination faster than cofactor expansion for larger matrices.

Worked elimination check

For the worked 3 by 3 matrix above, apply $R_3 \leftarrow R_3 + R_1$. Row replacement leaves the determinant unchanged and produces a triangular matrix with diagonal entries 1, 2, and 4. Therefore,

$$\det(A) = 1(2)(4) = 8$$

This agrees with the cofactor expansion. Using two independent methods is a useful arithmetic check while learning determinant rules.

Determinant and Invertibility

For a square matrix A , the following statements are equivalent:

- $\det(A) \neq 0$;
- A is invertible;
- $\text{rank}(A)$ equals the number of rows;
- $Ax = 0$ has only the trivial solution;
- $Ax = b$ has a unique solution for every compatible-size vector b .

For a 2 by 2 matrix,

$$A^{-1} = \frac{1}{\det(A)} \times \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

when $\det(A) \neq 0$.

This formula is useful for understanding, but solving a numerical system directly is usually preferable to explicitly forming an inverse.

Limits of the Inverse Claim

The determinant is defined here only for square matrices. A rectangular matrix has no ordinary determinant, even though it can still have rank and can define a meaningful linear system.

Also, $\det(A) = 0$ means a square matrix is noninvertible. It does not mean every associated system has no solution. Depending on b , the system can be inconsistent or can have infinitely many solutions.

Real Application: Area Change in an Image

A two-dimensional graphics transformation can stretch, shear, reflect, or collapse an image. If its determinant is 2, every small area is doubled. If its determinant is -2 , area is doubled and orientation is reversed. If the determinant is 0, a two-dimensional region is flattened into a line or point, so the original coordinates cannot be uniquely recovered.

This gives a practical interpretation of invertibility: an invertible transformation preserves enough information to undo the mapping. A zero determinant signals irreversible dimensional collapse.

Efficient Hand Calculation

For a matrix larger than 3 by 3, recursive cofactor expansion creates many smaller determinants. Row reduction is usually more efficient:

- 1 Use row replacement operations freely because they do not change the determinant.
- 2 Track every row swap by changing the sign.
- 3 Track every row scaling by multiplying the determinant by the same factor.
- 4 Once the matrix is triangular, multiply the diagonal entries and account for the recorded changes.

This method connects determinants to the same elimination ideas used for solving linear systems.

Numerical Caution

Raw determinant magnitude alone cannot diagnose ill-conditioning because rescaling a matrix changes its determinant. For numerical work, inspect the condition number or singular values instead.

Computing a large determinant by recursive cofactor expansion is expensive. Stable numerical libraries use factorization-based methods.

What the Determinant Does Not Tell You

- It has no ordinary definition for a rectangular matrix.
- Its magnitude depends on coordinate units and scaling.
- A nonzero value proves exact invertibility but does not guarantee numerical stability.
- A zero value does not decide whether one singular system has no solution or infinitely many solutions; the augmented rank decides that.
- It is a global summary and does not identify which rows or columns create dependence.

Common Mistakes

- Using $ad + bc$ instead of $ad - bc$ for a 2 by 2 determinant.

- Forgetting the alternating cofactor signs.
- Applying a row operation without accounting for its determinant effect.
- Treating determinant as an entrywise operation.
- Speaking of the determinant of a rectangular matrix.
- Concluding “no solution” from singularity without checking the augmented system.

Takeaway

The determinant connects algebra and geometry. It detects whether a square transformation preserves full dimension, measures signed scale, and links directly to invertibility and unique solvability.

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